



October 13, 2010

Bridgewater State University
131 Summer Street
Bridgewater, MA 02325

Dear Search Committee Members,

I am applying for a tenure-track position as posted on the AMS job listings website. I am a graduate student at the University of Kansas under the direction of Milena Stanislavova. Currently, I am completing a Ph.D. in Mathematics and I expect to graduate by May of 2011. I believe that my strong teaching background and my complete commitment to research make me an excellent candidate for your position.

I will bring experience and dedication that will compliment the educational goals of Bridgewater State University. I was the sole instructor for nine semesters at the University of Kansas and a teaching assistant for two semesters at Bogazici University in Turkey. I have taught several courses including Calculus, Vector Calculus and Differential Equations. Furthermore, I have been actively involved in community outreach programs and events for elementary and junior high school students, such as Math Awareness Month, the International Math Kangaroo competition, and the Math Olympiads and I hope to lead similar programs in Bridgewater.

I am interested in the long time behavior and the stability of special solutions of nonlinear partial differential equations. The methods that I use widely range from spectral and harmonic analysis to dynamical systems. I have obtained results on the long time behavior of the solutions of Kuramoto-Sivashinsky and Burger-Sivashinsky equations and stability results on Klein-Gordon type equations. The results on long time behavior have been published in *Journal Dynamics of PDEs*, *Discrete and Continuous Dynamical Systems* and *AIP Conference Proceedings*. Additionally, the stability results about the Klein-Gordon equation have recently been submitted.

I have enclosed my curriculum vitae, research statement, and teaching statement. You may also find my publications and preprints on my webpage at <http://www.math.ku.edu/~ademirkaya>. I will be attending at the joint meetings in New Orleans, LA. I have registered with the employment center and I am presenting a poster in the AWM Workshop. Thank you for your time and consideration. I look forward to hearing from you soon.

Sincerely,

Aslihan Demirkaya